

Lighthouse Partial Column Dissemination Comparison Report

Test Date: June 22, 2025
Lighthouse Version: v7.1.0-beta.0
Test Environment: Kurtosis + 4 CL nodes + 4 EL nodes

Executive Summary

This report compares the network performance and resource utilization of Lighthouse when partial column dissemination is **enabled** versus **disabled**. The tests involved identical stress testing scenarios with 50 transactions sent to each network configuration.

Test Configuration Comparison

Parameter	Enabled Test	Disabled Test
enable_partial_column_dissemination	true (default)	false
partial_column_cells_per_message	16 (default)	16
Docker Image	lighthouse-partial-columns:latest	lighthouse-partial-columns-disabled:latest
Network Participants	4 CL + 4 EL nodes	4 CL + 4 EL nodes

Transactions Sent	50	50
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Network Traffic Analysis

Discovery Protocol Traffic

Node	Enabled (bytes)		Disabled (bytes)		Difference
	Inbound	Outbound	Inbound	Outbound	Total Change
CL-1	947,693	819,158	131,928	156,525	-83.7%
CL-2	930,934	845,278	127,885	160,450	-83.8%
CL-3	477,769	955,094	156,592	125,862	-80.3%
CL-4	1,012,115	748,981	155,689	129,257	-83.8%
Total	3,368,511	3,368,511	572,094	572,094	-83.0%

Gossip Protocol Message Analysis

Message Type	Enabled Count	Disabled Count	Difference
Sync Committee Messages	~29,712	~12,251	-58.8%
Beacon Attestations	~400+	~44	-89.0%
Beacon Blocks	~232	~39	-83.2%
Sync Committee Contributions	Not measured	~249	N/A

Beacon Aggregate Proofs	Not measured	~99	N/A
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Performance Metrics

Partial Columns ENABLED

- **Total Network Traffic:** 6.42 MB
- **Average per Node:** 1.6 MB
- **Blocks Produced:** 62+ blocks
- **Peer Connections:** 2 per node (8 total)
- **Discovery Traffic:** High (3.21 MB inbound + 3.21 MB outbound)

Partial Columns DISABLED

- **Total Network Traffic:** 1.09 MB
- **Average per Node:** 0.27 MB
- **Blocks Produced:** 6+ blocks (1 during test)
- **Peer Connections:** 2 per node (8 total)
- **Discovery Traffic:** Low (0.55 MB inbound + 0.55 MB outbound)

Key Findings

1. Network Traffic Reduction

Disabling partial column dissemination resulted in a **83% reduction** in total network traffic:

- Enabled: 6.42 MB total traffic
- Disabled: 1.09 MB total traffic
- Savings: 5.33 MB (83.0% reduction)

2. Gossip Message Efficiency

Significant reductions in gossip protocol messages:

- Sync committee messages: 58.8% reduction
- Beacon attestations: 89.0% reduction
- Beacon blocks: 83.2% reduction

3. Block Production Impact

The enabled version produced significantly more blocks during the test period, suggesting more active consensus participation.

4. Peer Connectivity

Both configurations maintained identical peer connectivity (2 peers per node), indicating that the feature does not affect basic network topology.

Technical Analysis

Why the Difference?

The partial column dissemination feature appears to increase network activity by:

- **Enhanced Gossip Propagation:** More aggressive message broadcasting for data availability
- **Increased Discovery Activity:** More frequent peer discovery and maintenance
- **Data Column Preparation:** Pre-processing for future PeerDAS features
- **Consensus Synchronization:** More frequent sync committee participation

Code Change Impact:

```
enable_partial_column_dissemination: false
```

This single boolean flag controls the activation of partial column dissemination logic throughout the Lighthouse codebase, affecting gossip handling, data

column processing, and network synchronization patterns.

Resource Utilization

Resource	Enabled	Disabled	Efficiency Gain
Network Bandwidth	6.42 MB	1.09 MB	83% reduction
Discovery Protocol	High activity	Low activity	83% reduction
Gossip Messages	~30,344 messages	~12,633 messages	58% reduction
Peer Connections	8 total	8 total	No change

Implementation Details

The partial column dissemination implementation is available at:

<https://github.com/gitToki/lighthouse/tree/partial-column-d>

This implementation demonstrates the impact of partial column dissemination on network resource utilization, showing an 83% reduction in network traffic when the feature is disabled.

Test Environment Details

Configuration: 4 Lighthouse CL nodes + 4 Geth EL nodes
Network: Kurtosis-managed private testnet Genesis:
Mainnet preset with custom fork schedule Test Duration:
~5 minutes per configuration Stress Test: 50 identical
transactions per configuration Enabled Build:
enable_partial_column_dissemination = true (default)
Disabled Build: enable_partial_column_dissemination =
false Data Collection: Prometheus metrics from each CL
node Analysis Period: Post-deployment steady state

Report generated on June 22, 2025 using automated testing framework.